Data Analytics

Exercise 1

SQL Fundamentals

Employees

id	first_name	last_name	Department	Salary	hire_date	City
1	John	Doe	IT	55 000	2019-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johson	Finance	60 000	2017-09-30	Los_Angele
4	Sarah	Brown	IT	53000	2021-03-25	NY
5	David	White	Marketing	52 000	2016-04-10	San Francisco
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston
8	Jessica	Moore	HR	51 000	2018-05-22	LA
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	SF

Questions

(1) SELECT STATEMENT

Write a SQL query to retrieve all columns from employee table

SELECT * FROM employees_table;

Table will be the same as employees_table

(2) SELECT DISTINCT STATEMENT → Find all unique departments

SELECT DISTINCT departments
FROM employees table;

Department
IT
HR
Finance
Marketing

③ ORDER BY STATEMENT

Write SQL retrieve all employees first & last names ordered by salary in descending order

```
SELECT first name,
       Last name,
       Salary
  FROM employees
  ORDER BY salary DESC
```

First name	Last name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000

④ LIMIT Statement

Write SQL query to retrieve the top 5 highest paid employees

```
SELECT * First name,
         Last name,
         Salary
FROM employees
ORDER BY salary DESC
  LIMIT 5;
```

First name	Last name	Salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Willson	59000

5 WHERE Statement
SQL Query find employees who work in the IT department

SELECT id
first name
Lastname
Department
FROM employees
WHERE 'IT' department;

First name	Lastname	Department
John	Doe	IT
Sarah	Brown	IT

6 AND Statement
Write SQL query to find employees in the finance department AND have a salary greater than 58000

SELECT Firstname
Lastname
Salary
FROM employees
WHERE department = 'finance' AND Salary > 58000;

first name	Lastname	Department	Salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59000

7 OR Statement
Write SQL query to find employees who work in HR depart
OR Marketing

```
SELECT First name
       Last name
FROM employees
WHERE department =' HR or Department', = 'Marketing';
```

First name	Last name	Department
Jane	Smith	HR
Jessica	Moore	HR
David	White	Marketing

⑧ NOT STATEMENT

Write SQL query to find employees who do not work in IT department

```
SELECT Firstname
       Lastname
FROM employees
WHERE departmes NOT 'IT department'
```

↳ WHERE NOT department = 'IT'

⑨ IN STATEMENT

find employees who are in the HR, IT or finance department

```
SELECT first name
       Lastname
FROM employees
WHERE department IN (' HR, IT' finance');
```

(10) Combining Conditions
WRite SQL query to find employes who are in IT depart ment thare salary greater than 50000 and Located in New YORK

```
SELECT First name
       Last name
       Salary  city
FROM employees
WHERE department = 'IT'
  AND Salary > 50000
  AND city = 'New york'
```

Fistname	Last name	Dept	Salary	City
John	Doe	IT	55000	New York

(11) Combining WHERE AND and ORDER BY
Write SQL query to retrieve the First & last names of employees who work in the Finance or Marketing department earn more than 52 000 and order the results by salary in descending order

```
SELECT First name
       Last name
       Salary
FROM employees department = 'Finance or Marketing AND
WHERE Salary > 52 000
ORDER BY Salary DESC;
```

```
SELECT First name
       Last name
FROM employees
WHERE department = 'HR
```

First name	Last name
Jane	Smith
Jessica	Moore
David	White

(8) NOT STATEMENT
Write SQL query to
IT departme

```
SELECT Firstname
       Lastnam
FROM employee
WHERE departme
```
WHERE NO

(9) IN STATEMENT
find employ

```
SELECT Fi
       L
FROM e
WHERE
```

(10) Combining Conditions

WRite SQL query to find employes who are in IT department there salary greater than 50000 and Located in New YORK

```
SELECT first name
       Last name
       Salary city
FROM employees
WHERE department = 'IT'
  AND Salary > 50000
  AND city = 'New york'
```

Fistname	Last name	Dept	Salary	City
John	Doe	IT	55000	New York

(11) Combining WHERE AND and ORDER BY

Write SQL query to retrieve the First & last names of employees who work in the Finance or Marketing department earn more than 52 000 and order the results by salary in descending order

```
SELECT first name
       Last name
       Salary
FROM employees
department = 'Finance or Marketing AND
WHERE Salary > 52 000
ORDER BY salary DESC;
```

(12) Combining SELECT DISTINCT, WHERE and IN

Write SQL find all the unique cities where employees work, excluding those in IT and HR

```
SELECT DISTINCT cities
FROM employees
WHERE department NOT IN('IT','HR');
```

(13) Combining WHERE NOT AND ORDER BY

Write SQL query to retrieve employes who are NOT in the Finance department, have salary > 50000 & order the results by hire date in ascending order.

```
SELECT first name
       Last name
       Department
       Salary
       Hire_date
FROM employees
WHERE NOT 'Department'='Finance'
AND Salary > 50000
ORDER BY ASC;
```

(14) Combining WHERE OR, IN And LIMIT

Write a SQL to find 3 employees who work either Chicago or LA and belong to the IT or Marketing

```
SELECT first name
       Last name
       City
FROM employees
WHERE department 'IT or Department'='Marketing
AND IN Chicago or LA
LIMIT 3;
```

(12) Combining SELECT DISTINCT, WHERE and IN

Write SQL find all the unque cities where employees work, excluding those in IT and HR

```
SELECT DISTINCT cities ;
  FROM employees
 WHERE department NOT IN('IT','HR');
```

(13) Combining WHERE NOT AND ORDER BY

Write SQL query to retrieve employes who are NOT in the Finance department, have salary > 50000 & order the results by hire date in ascending order.

```
SELECT First name
       Last name
       Department
       Salary
       Hire_date
FROM employees
WHERE NOT Department'='Finance'
AND Salary > 50000
ORDER BY ASC;
```

(14) Combining WHERE OR, IN And LIMIT

Write a SQL to find 3 employees who work either Chicago or LA and belong to the IT or Marketing

```
SELECT First name
       Last name
       City
FROM employees
WHERE department 'IT or Department'='Marketing
AND IN Chicago or LA
LIMIT 3;
```

```
SELECT First name        ← Corrections
       Last name
       Department
       City
FROM employees
WHERE City IN('Chicago','LA')
AND Department IN ('IT','Marketing')
LIMIT 3;
```

15 Combining WHERE AND OR NOT ORDER BY and LIMIT

```
Select FIRST name
       Last name
       Department
       Salary
       City
FROM employees
WHERE (Depatment = IT or departme = 'finance')
AND city 'San Francisco'
AND Salary > 55000
ORDER BY salary DESC
LIMIT 5;
```